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# Safety data sheet



BASF Safety data sheet according to Regulation UK SI 2019/758 and UK SI 2020/1577 as amended from time to time.
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Product: **DisPlus 39L**
(ID no. 30735595/SDS_GEN_GB/EN)
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## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

## DisPlus 39L

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Performance Chemicals for Oilfield Applications

For the detailed identified uses of the product see appendix of the safety data sheet.

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF plc
4th and 5th Floors, 2 Stockport Exchange
Railway Road, Stockport, SK1 3GG
UNITED KINGDOM

Telephone: +44 161 475 3000
E-mail address: product-safety-uk-and-ireland@basf.com

### 1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## SECTION 2: Hazards Identification

### 2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

| | |
|---|---|
| Acute Tox. 4 (oral) | H302 Harmful if swallowed. |
| Acute Tox. 4 (dermal) | H312 Harmful in contact with skin. |
| Skin Corr./Irrit. 2 | H315 Causes skin irritation. |
| Eye Dam./Irrit. 1 | H318 Causes serious eye damage. |
| Aquatic Chronic 3 | H412 Harmful to aquatic life with long lasting effects. |

For the classifications not written out in full in this section the full text can be found in section 16.

## 2.2. Label elements

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H318 | Causes serious eye damage. |
| H315 | Causes skin irritation. |
| H302 + H312 | Harmful if swallowed or in contact with skin. |
| H412 | Harmful to aquatic life with long lasting effects. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P273 | Avoid release to the environment. |

Precautionary Statements (Response):

| | |
|---|---|
| P310 | Immediately call a POISON CENTER or physician. |
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P303 + P352 | IF ON SKIN (or hair): Wash with plenty of soap and water. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Hazard determining component(s) for labelling: Alcohols, C12-14, ethoxylated, sulfates, sodium salts, Alcohols, C13-15, branched and linear, ethoxylated, 2-butoxyethanol

## 2.3. Other hazards

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Not applicable

### 3.2. Mixtures

Chemical nature

Preparation based on: anionic and non-ionic surfactants

Hazardous ingredients (GHS)

Alcohols, C12-14, ethoxylated, sulfates, sodium salts

| | |
|---|---|
| Content (W/W): >= 25 % - < 50 % | Skin Irrit. 2 |
| CAS Number: 68891-38-3 | Eye Dam. 1 |
| EC-Number: 500-234-8 | H318, H315 |
| REACH registration number: 01-2119488639-16 | |

Alcohols, C13-15, branched and linear, ethoxylated

| | |
|---|---|
| Content (W/W): >= 25 % - < 50 % | Acute Tox. 4 (oral) |
| CAS Number: 157627-86-6 | Eye Dam./Irrit. 1 |
| | Aquatic Chronic 3 |
| | H318, H302, H412 |

2-butoxyethanol

| | |
|---|---|
| Content (W/W): >= 20 % - < 25 % | Acute Tox. 4 (Inhalation - vapour) |
| CAS Number: 111-76-2 | Acute Tox. 4 (oral) |
| EC-Number: 203-905-0 | Skin Irrit. 2 |
| REACH registration number: 01-2119475108-36 | Eye Irrit. 2 |
| INDEX-Number: 603-014-00-0 | H319, H315, H302 + H332 |

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote, administer corticosteroid dose aerosol to prevent pulmonary odema.

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam

### 5.2. Special hazards arising from the substance or mixture

Endangering substances: harmful vapours, carbon oxides
Advice: Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

### 5.3. Advice for fire-fighters
Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

## SECTION 6: Accidental Release Measures

### 6.1. Personal precautions, protective equipment and emergency procedures
Use personal protective clothing. Breathing protection required. Avoid contact with skin and eyes. Avoid inhalation. Ensure adequate ventilation.

### 6.2. Environmental precautions
Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

### 6.3. Methods and material for containment and cleaning up
For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

### 6.4. Reference to other sections
Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling
Wear chemically resistant gloves in combination with intensive management supervision control. Use suitable eye protection. Local exhaust ventilation Wear suitable respiratory protection.

No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Remove contaminated clothing and protective equipment before entering eating areas.

Protection against fire and explosion:
Take precautionary measures against static discharges.

### 7.2. Conditions for safe storage, including any incompatibilities
Segregate from foods and animal feeds. Segregate from strong acids. Segregate from strong bases.
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

### 7.3. Specific end use(s)
See exposure scenario(s) in the attachment to this safety data sheet.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

111-76-2: 2-butoxyethanol
Skin Designation (WEL/EH 40 (UK))
The substance can be absorbed through the skin.
TWA value 123 mg/m3 ; 25 ppm (WEL/EH 40 (UK))
Skin Designation (OEL (EU))
The substance can be absorbed through the skin.
STEL value 246 mg/m3 ; 50 ppm (OEL (EU))
indicative
TWA value 98 mg/m3 ; 20 ppm (OEL (EU))
indicative
STEL value 246 mg/m3 ; 50 ppm (WEL/EH 40 (UK))
Ceiling limit value/factor: 15 min
STEL value 246 mg/m3 ; 50 ppm (EU SCOEL)
Ceiling limit value/factor: 15 min
TWA value 98 mg/m3 ; 20 ppm (EU SCOEL)
Ceiling limit value/factor: 8HR
Skin Designation (EU SCOEL)
The substance can be absorbed through the skin.

Components with biological limit values

111-76-2: 2-butoxyethanol
UKEH40BMGV
Determinant: butoxyacetic acid
Biological Specimen: Creatinine in urine
Sampling time: End of shift
Concentration: 240 mmol/mol

PNEC
Data refer to the lead substance

Components with PNEC

68891-38-3: Alcohols, C12-14, ethoxylated, sulfates, sodium salts
freshwater: 0.129 mg/l
marine water: 0.0129 mg/l
intermittent release: 0.71 mg/l
STP: 10000 mg/l
sediment (freshwater): 4.835 mg/kg
sediment (marine water): 0.4835 mg/kg
soil: 7.5 mg/kg

oral (secondary poisoning):
No PNEC oral derived, as accumulation in organisms is not to be expected.

DNEL
Data refer to the lead substance

Components with DNEL

111-76-2: 2-butoxyethanol
worker: Long-term exposure- systemic effects, Inhalation: 98 mg/m3, 20 ppm
worker: Short-term exposure - systemic effects, Inhalation: 1091 mg/m3
worker: Short-term exposure - local effects, Inhalation: 246 mg/m3
consumer: Long-term exposure- systemic effects, oral: 6.3 mg/kg
consumer: Short-term exposure - systemic effects, oral: 26.7 mg/kg
consumer: Long-term exposure- systemic effects, Inhalation: 59 mg/m3
consumer: Short-term exposure - systemic effects, Inhalation: 426 mg/m3
consumer: Short-term exposure - local effects, Inhalation: 147 mg/m3

## 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Wear respiratory protection if ventilation is inadequate. Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
nitrile rubber (NBR) - 0.4 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Wearing of closed work clothing is required additionally to the stated personal protection equipment. Take off immediately all contaminated clothing. Wash contaminated clothing before reuse. Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

Do not discharge product into the environment without control.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| Form: | liquid | |
| Colour: | clear | |
| Odour: | product specific | |
| Odour threshold: | not determined | |
| pH value: | > 5.5<br>(100 %(m)) | (DIN 19268) |
| freezing range: | not determined | |
| boiling temperature: | 100 - 101 °C<br>(1,013 hPa) | (OECD Guideline 103) |
| Flash point: | 100 - 101 °C | (ISO 2719) |
| Evaporation rate: | not determined | |
| Flammability: | hardly combustible | (derived from flash point) |
| Lower explosion limit: | For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point. | |
| Upper explosion limit: | For liquids not relevant for classification and labelling. | (DIN 51649-1) |
| Ignition temperature: | 232 °C<br>The product has not been tested.<br>The statement has been derived from the properties of the individual components. | |
| *Information on: 2-butoxyethanol* | | |
| *Ignition temperature:* | *232 °C* | *(DIN EN 14522)* |

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| | | |
|---|---|---|
| Vapour pressure: | 62.74 hPa<br>(20 °C) | (OECD Guideline 104) |
| | 71.71 hPa<br>(50 °C) | (OECD Guideline 104) |
| | 155.82 hPa<br>(55 °C) | (OECD Guideline 104) |
| Density: | 1.02 g/cm3<br>(15 °C) | (ISO 2811-3) |
| | 1.02 g/cm3<br>(20 °C) | (ISO 2811-3) |
| | 1.00 g/cm3<br>(50 °C) | |

Relative vapour density (air): not determined
Solubility in water: soluble
Solubility (qualitative) solvent(s): alcohols soluble
Partitioning coefficient n-octanol/water (log Kow): not applicable for mixtures
Self ignition: not self-igniting

Thermal decomposition: not determined
Viscosity, dynamic: 60 mPa.s (23 °C)
Viscosity, kinematic: approx. 59 mm2/s (23 °C) (calculated (from dynamic viscosity))
Explosion hazard: not explosive
Fire promoting properties: not fire-propagating

### 9.2. Other information

Self heating ability: It is not a substance capable of spontaneous heating.

Hygroscopy: hygroscopic
Grain size distribution: The substance / product is marketed or used in a non solid or granular form.

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

### 10.2. Chemical stability
The product is stable if stored and handled as prescribed/indicated.

### 10.3. Possibility of hazardous reactions
No hazardous reactions when stored and handled according to instructions.

### 10.4. Conditions to avoid
Avoid open flames.

### 10.5. Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

### 10.6. Hazardous decomposition products

Hazardous decomposition products:

No hazardous decomposition products known.

## SECTION 11: Toxicological Information

### 11.1. Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after short-term skin contact. Of moderate toxicity after single ingestion.

Experimental/calculated data:
ATE (oral): > 300 - < 2,000 mg/kg (calculated)

ATE (by inhalation): 50 mg/l

ATE (dermal): 2,000 mg/kg

*Information on: 2-butoxyethanol*
*Assessment of acute toxicity:*
*Of moderate toxicity after single ingestion. The inhalation of a highly enriched/saturated vapor-air-mixture represents an unlikely acute hazard. Virtually nontoxic after a single skin contact.*
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*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*

*Experimental/calculated data:*
*LD50 rat (oral): > 2,000 mg/kg (BASF-Test)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*

*LD50 rat (oral): > 2,000 mg/kg (BASF-Test)*
*The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*

*Experimental/calculated data:*
*LD50 rat (oral): > 300 - 2,000 mg/kg (OECD Guideline 401)*
*Literature data.*

*Information on: 2-butoxyethanol*

*Experimental/calculated data:*
*LD50 guinea pig (oral): 1,200 mg/kg (similar to OECD guideline 401)*

*ATE (oral): 1,200 mg/kg*
---

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Experimental/calculated data:*
*LD50 rat (dermal): 2,000 mg/kg (OECD Guideline 402)*
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Irritation

Assessment of irritating effects:
Skin contact causes irritation. May cause severe damage to the eyes.

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Experimental/calculated data:*
*Skin corrosion/irritation*
*In vitro assay: Irritant. (OECD Guideline 439)*

*Skin corrosion/irritation*
*rabbit: Irritant. (OECD Guideline 404)*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Experimental/calculated data:*
*Skin corrosion/irritation*
*rabbit: non-irritant (OECD Guideline 404)*
*Literature data.*

*Information on: 2-butoxyethanol*
*Experimental/calculated data:*
*Skin corrosion/irritation*
*rabbit: Irritant. (BASF-Test)*
---

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Experimental/calculated data:*
*Serious eye damage/irritation*
*In vitro assay: no irreversible damage (BCOP)*

*Serious eye damage/irritation*
*rabbit: irreversible damage (OECD Guideline 405)*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Experimental/calculated data:*
*Serious eye damage/irritation*
*rabbit: irreversible damage (Draize test)*
*Literature data.*

*Information on: 2-butoxyethanol*
*Experimental/calculated data:*
*Serious eye damage/irritation*
*rabbit: Irritant. (OECD Guideline 405)*
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Respiratory/Skin sensitization

Assessment of sensitization:
Based on available data, the classification criteria are not met.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
Acutely toxic for aquatic organisms. Harmful to aquatic life with long lasting effects.

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Toxicity to fish:*
*LC50 (96 h) 7.1 mg/l, Brachydanio rerio (OECD Guideline 203, Flow through.)*
*The details of the toxic effect relate to the nominal concentration.*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Toxicity to fish:*
*LC50 (96 h) > 1 - 10 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 84/449/EWG, C.1, static)*
*Literature data.*
---

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Aquatic invertebrates:*
*EC50 (48 h) 7.4 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)*
*The details of the toxic effect relate to the nominal concentration.*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Aquatic invertebrates:*
*EC50 (48 h) > 1 - 10 mg/l, Daphnia magna (Directive 79/831/EEC)*
*Literature data.*
---

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Aquatic plants:*
*No observed effect concentration (72 h) 0.95 mg/l (growth rate), Scenedesmus subspicatus (Algal growth inhibition test)*
*The statement of the toxic effect relates to the analytically determined concentration.*

*EC50 (72 h) 27.7 mg/l (growth rate), Desmodesmus subspicatus (OECD Guideline 201)*
*The statement of the toxic effect relates to the analytically determined concentration.*

*EC10 (72 h) 4.4 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)*
*The statement of the toxic effect relates to the analytically determined concentration.*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Aquatic plants:*
*EC50 (72 h) > 1 - 10 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)*
*Literature data.*

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*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Chronic toxicity to fish:*
*EC10 (28 d) 1.7 mg/l, Pimephales promelas (OECD Guideline 210, Flow through.)*
*The statement of the toxic effect relates to the analytically determined concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
---

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Chronic toxicity to aquatic invertebrates:*
*No observed effect concentration (21 d) > 1.19 mg/l, Daphnia magna (OECD Guideline 211, other)*
*The statement of the toxic effect relates to the analytically determined concentration.*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Chronic toxicity to aquatic invertebrates:*
*No observed effect concentration > 0.1 - 1 mg/l, Daphnia magna*
*Literature data.*
---

## 12.2. Persistence and degradability

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Assessment biodegradation and elimination (H2O):*
*Readily biodegradable (according to OECD criteria).*

*Information on: 2-butoxyethanol*
*Assessment biodegradation and elimination (H2O):*
*Readily biodegradable (according to OECD criteria).*
---

## 12.3. Bioaccumulative potential

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Assessment bioaccumulation potential:*
*Significant accumulation in organisms is not to be expected.*

*Information on: Alcohols, C13-15, branched and linear, ethoxylated*
*Assessment bioaccumulation potential:*
*Accumulation in organisms is not to be expected.*

*Information on: 2-butoxyethanol*
*Assessment bioaccumulation potential:*
*Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.*
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## 12.4. Mobility in soil

*Information on: Alcohols, C12-14, ethoxylated, sulfates, sodium salts*
*Assessment transport between environmental compartments:*
*Volatility: The substance will not evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.Study scientifically not justified.*

*Information on: 2-butoxyethanol*
*Assessment transport between environmental compartments:*
*Volatility: The substance will not evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Adsorption to solid soil phase is not expected.*
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### 12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

### 12.6. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### 12.7. Additional information

Other ecotoxicological advice:
The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

The UK Environmental Protection (Duty of Care) Regulations (EP) and amendments should be noted (United Kingdom).

This product and any uncleaned containers must be disposed of as hazardous waste in accordance with the 2005 Hazardous Waste Regulations and amendments (United Kingdom)

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

RID

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |

| | |
|---|---|
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

UK REACH SI, Annex XVII, Marketing and Use Restrictions
Number on List: 3

UK REACH SI, Annex XVII, Marketing and Use Restrictions
Number on List: 54

UK REACH SI, Annex XVII, Marketing and Use Restrictions
Number on List: 55

2-(2-methoxyethoxy)ethanol

2-(2-butoxyethoxy)ethanol

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
Listed in above regulation: no
Classification applies for standard conditions of temperature and pressure.

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

The data should be considered when making any assessment under the Control of Substances Hazardous to Health Regulations (COSHH), and related guidance, for example, 'COSHH Essentials' (United Kingdom).

### 15.2. Chemical Safety Assessment

Assessment of safe use has been performed for the mixture. Results and measures for human health are documented in sections 7 and 8, those for environment are attached as an annex of the safety data sheet.

## SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Acute Tox. 4 (oral)
Acute Tox. 4 (dermal)
Acute Tox. 5 (Inhalation - vapour)

Eye Dam./Irrit. 1
Skin Corr./Irrit. 2
Aquatic Acute 2
Aquatic Chronic 3

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Acute Tox. | Acute toxicity |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| Skin Irrit. | Skin irritation |
| Eye Dam. | Serious eye damage |
| Eye Irrit. | Eye irritation |
| H318 | Causes serious eye damage. |
| H315 | Causes skin irritation. |
| H302 + H312 | Harmful if swallowed or in contact with skin. |
| H412 | Harmful to aquatic life with long lasting effects. |
| H302 | Harmful if swallowed. |
| H319 | Causes serious eye irritation. |
| H302 + H332 | Harmful if swallowed or if inhaled. |

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

## Annex: Exposure Scenarios

**Index**

**1.** Formulation of mixtures (industrial, professional, consumer)
SU0-1; ERC2

**2.** End use (industrial, professional)
SU5, SU12, SU13, SU15; ERC4

**3.** End use (industrial, professional)
SU0-1, SU8, SU9; ERC6a

**4.** End use (industrial, professional)
SU5, SU12, SU13, SU15; ERC6b

**5.** End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8a

**6.** End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8b

**7.** End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8d

**8.** End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8e

**********

### 1. Short title of exposure scenario

Formulation of mixtures (industrial, professional, consumer)
SU0-1; ERC2
A consolidation approach has been used to limit the number of exposure scenarios. Parameters provided in the scenario cover all mentioned spERCs.

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | AISE SPERC 2.1.a.v2: AISE SPERC 2.1.a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 250 |
| Emission factor air | 0 % |
| Emission factor water | 0.01 % |
| Emission factor soil | 0 % |

| | |
|---|---|
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Wastewater treatment measures considered suitable are, e.g. | Precipitation, Coagulation, Must be eliminated from water by chemical flocculation. |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 99,810 kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | CEPE SPERC 2.1a.v1: CEPE SPERC 2.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 225 |
| Emission factor air | 0.6 % |
| Emission factor water | 0 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 273,665 kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | COLIPA SPERC 2.1.a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 250 |
| Emission factor air | 0 % |
| Emission factor water | 0.1 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Wastewater treatment measures considered suitable are, e.g. | Sedimentation, Precipitation, pH adjustment |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 78,947 kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | EFCC SPERC 2.1a.v1: EFCC SPERC 2.1a.v1 ; covers also: EFCC SPERC 2.1b.v1: EFCC SPERC 2.1b.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |
| Emission factor air | 1 % |
| Emission factor water | 0.5 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |

| | |
|---|---|
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 43,802 kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC2: Formulation into mixture |
| ***Operational conditions*** | |
| Minimum emission days per year | 100 |
| Emission factor air | 0.25 % |
| Emission factor water | 0.3 % |
| Emission factor soil | 0.01 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 52,076 kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 2.2.v1: ESVOC SpERC 2.2.v1 ; covers also: ESVOC SpERC 1.1b.v1: ESVOC SpERC 1.1b.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 300 |

| | |
|---|---|
| Emission factor air | 0.25 % |
| Emission factor water | 0.5 % |
| Emission factor soil | 0.01 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Wet scrubber - for dusts, Filtration, Waste gas treatment by thermal oxidation, Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment, Distillation |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 157,188 kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | FEICA SPERC 2.1a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |
| Emission factor air | 1 % |
| Emission factor water | 0.005 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |

| | |
|---|---|
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 179,837 kg/d |
| Data refer to the product | |

**********

## 2. Short title of exposure scenario

End use (industrial, professional)
SU5, SU12, SU13, SU15; ERC4
A consolidation approach has been used to limit the number of exposure scenarios. Parameters provided in the scenario cover all mentioned spERCs.

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | AISE SPERC 4.1.v2: AISE SPERC 4.1.v2 ; covers also: AISE SPERC 5.1b.v2: AISE SPERC 5.1b.v2 AISE SPERC 5.1a.v2: AISE SPERC 5.1a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |
| Emission factor air | 0 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 161 kg/d |

| | |
|---|---|
| | |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | CEPE SPERC 4.1a.v1: CEPE SPERC 4.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 225 |
| Emission factor air | 0.8 % |
| Emission factor water | 2 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Waste gas treatment by thermal oxidation |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 4,520 kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | EFCC SPERC 4.1a.v1: EFCC SPERC 4.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |
| Emission factor air | 98.5 % |
| Emission factor water | 0 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |

| | |
|---|---|
| | |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 12,315 kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 4.20.v1: ESVOC SpERC 4.20.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 300 |
| Emission factor air | 0.2 % |
| Emission factor water | 1 % |
| Emission factor soil | 0.01 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 12,797 kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | FEICA SPERC 4.2a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |

| | |
|---|---|
| | |
| Emission factor air | 98.5 % |
| Emission factor water | 0 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 3,694<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | TEGEWA SPERC 4.1.v1: TEGEWA SPERC 4.1.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |
| Emission factor air | 0 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |

| | |
|---|---|
| Maximum amount of safe use | 161<br>kg/d |
| Data refer to the product | |

**********

## 3. Short title of exposure scenario

End use (industrial, professional)
SU0-1, SU8, SU9; ERC6a

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC6a: Use of intermediate |
| ***Operational conditions*** | |
| Minimum emission days per year | 300 |
| Emission factor air | 0.025 % |
| Emission factor water | 0.05 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 4,080<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ESVOC SpERC 6.1a.v1: ESVOC SpERC 6.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 300 |
| Emission factor air | 2 ppm |

| | |
|---|---|
| | |
| Emission factor water | 1 % |
| Emission factor soil | 0.1 % |
| | Values provided in per mill |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Air treatment measures considered suitable are, e.g. | Wet scrubber - for dusts, Adsorption |
| Wastewater treatment measures considered suitable are, e.g. | Acclimated biological treatment |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 24,039 kg/d |
| Data refer to the product | |

**********

## 4. Short title of exposure scenario

End use (industrial, professional)
SU5, SU12, SU13, SU15; ERC6b

## Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | TEGEWA SPERC 6b.1.v1: TEGEWA SPERC 6b.1.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 220 |
| Emission factor air | 0 % |
| Emission factor water | 2 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |

| | |
|---|---|
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 10,574 kg/d |
| Data refer to the product | |

**********

## 5. Short title of exposure scenario

End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8a

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | AISE SPERC 8a.1.b.v2: AISE SPERC 8a.1.b.v2 ; covers also: AISE SPERC 8a.1.a.v2: AISE SPERC 8a.1.a.v2 AISE SPERC 8a.1.c.v2: AISE SPERC 8a.1.c.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine |

| | |
|---|---|
| | sediment. |
| Maximum amount of safe use | 216<br>kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | CEPE SPERC 8a.2a.v1: CEPE SPERC 8a.2a.v1 ; covers also: CEPE SPERC 8a.1a.v1: CEPE SPERC 8a.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 225 |
| Emission factor air | 99 % |
| Emission factor water | 1 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Wastewater treatment measures considered suitable are, e.g. | STP |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 153<br>kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8a: Widespread use of non-reactive processing aid (no inclusion into or onto article, indoor) |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 100 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |

| | |
|---|---|
| | |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 113<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | COLIPA SPERC 8a.1.a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 6.5<br>kg/d |
| Data refer to the product | |

| |
|---|
| **Contributing exposure scenario** |

| | |
|---|---|
| **Use descriptors covered** | EFCC SPERC 8A.1a.v1: EFCC SPERC 8A.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 98 % |
| Emission factor water | 1 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 67<br>kg/d |
| Data refer to the product | |

**********

## 6. Short title of exposure scenario

End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8b

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC8b: Widespread use of reactive processing aid (no inclusion into or onto article, indoor) |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0.1 % |
| Emission factor water | 2 % |
| Emission factor soil | 0 % |

| | |
|---|---|
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 167<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | AISE SPERC 8a.1.a.v2: AISE SPERC 8a.1.a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 9.9<br>kg/d |
| Data refer to the product | |

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## 7. Short title of exposure scenario

End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8d

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | AISE SPERC 8a.1.a.v2: AISE SPERC 8a.1.a.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0 % |
| Emission factor water | 100 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 19<br>kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8d: Widespread use of non-reactive processing aid (no inclusion into or onto article, outdoor) |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 100 % |
| Emission factor water | 100 % |
| Emission factor soil | 20 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |

| | |
|---|---|
| | |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 97<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ECPA SPERC 8d.2.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 100 % |
| Emission factor water | 0 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | no STP |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine water. |
| Maximum amount of safe use | 9.2<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ECPA SPERC 8d.1.v1: ECPA SPERC 8d.1.v1 |
| ***Operational conditions*** | |

| | |
|---|---|
| Minimum emission days per year | 365 |
| Emission factor air | 0 % |
| Emission factor water | 0 % |
| Emission factor soil | 100 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | no STP |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine water. |
| Maximum amount of safe use | 9.2<br>kg/d |
| Data refer to the product | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ECPA SPERC 8d.2.v2 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 100 % |
| Emission factor water | 0 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | no STP |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine water. |
| Maximum amount of safe use | 9.2 |

| | |
|---|---|
| | kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ECPA SPERC 8d.1.v1: ECPA SPERC 8d.1.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0 % |
| Emission factor water | 0 % |
| Emission factor soil | 100 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | no STP |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine water. |
| Maximum amount of safe use | 9.2<br>kg/d |
| Data refer to the product | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | EFCC SPERC 8D.1a.v1: EFCC SPERC 8D.1a.v1 |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 98 % |
| Emission factor water | 1 % |
| Emission factor soil | 0 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |

| | |
|---|---|
| | |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 101<br>kg/d |
| Data refer to the product | |

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## 8. Short title of exposure scenario

End use (industrial, professional)
SU5, SU11, SU12, SU13, SU15; ERC8e

### Control of exposure and risk management measures

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | ERC8e: Widespread use of reactive processing aid (no inclusion into or onto article, outdoor) |
| ***Operational conditions*** | |
| Minimum emission days per year | 365 |
| Emission factor air | 0.1 % |
| Emission factor water | 2 % |
| Emission factor soil | 1 % |
| Receive Surf. Water (Flow Rate). | 18,000 m3/d |
| Dilution factor river | 10 |
| Dilution factor coast | 100 |
| ***Risk Management Measures*** | |
| Type of STP | Municipal STP |
| Assumed sewage treatment plant flow (m3/d) | 2,000 m3/d |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | EASY TRA v4.1, ECETOC TRA v3.0, Environment |
| | Risk from environmental exposure is driven by marine sediment. |
| Maximum amount of safe use | 67<br>kg/d |

Data refer to the product

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